The Fundamental Theorem of Calculus

ANSWER KEY

Part 2: evaluating definite integrals

1 Evaluate each definite integral:

a $\int_1^3 (2x+1)\,dx$ $[x^2+x]_1^3 = 12 - 2 = 10.$

b $\int_0^\pi \sin x\,dx$ $[-\cos x]_0^\pi = 1-(-1) = 2.$

c $\int_1^4 \sqrt{x}\,dx$ $\left[\frac{2}{3}x^{3/2}\right]_1^4 = \frac{16}{3} - \frac{2}{3} = \frac{14}{3}.$

2 Evaluate $\int_0^1 e^{2x}\,dx$, to three decimal places.
$\left[\frac{1}{2}e^{2x}\right]_0^1 = \frac{1}{2}(e^2-1) \approx 3.195.$

Part 1: differentiating an integral

3 Use the Fundamental Theorem, Part 1, to find $\frac{d}{dx}\int_2^x (t^3+1)\,dt.$
$= x^3 + 1$ (directly, by FTC Part 1).

4 Find $\frac{d}{dx}\int_1^{x^2} \sin(t)\,dt$ using the chain rule with FTC.
$= \sin(x^2)\cdot 2x = 2x\sin(x^2).$

5 Let $g(x) = \int_0^x \sqrt{4+t^2}\,dt$. Find $g'(3)$.
$g'(x) = \sqrt{4+x^2}$, so $g'(3) = \sqrt{13}$.

Average value and applications

6 Find the average value of $f(x) = x^2$ on $[0, 3]$.
$\frac{1}{3}\int_0^3 x^2\,dx = \frac{1}{3}\cdot 9 = 3.$

7 A car's velocity is $v(t) = 3t^2 + 2$ m/s for $0 \le t \le 4$. Find the total distance traveled, and the average velocity over this interval.
Distance $= \int_0^4 (3t^2+2)\,dt = [t^3+2t]_0^4 = 64 + 8 = 72$ m. Average velocity $= \frac{72}{4} = 18$ m/s.